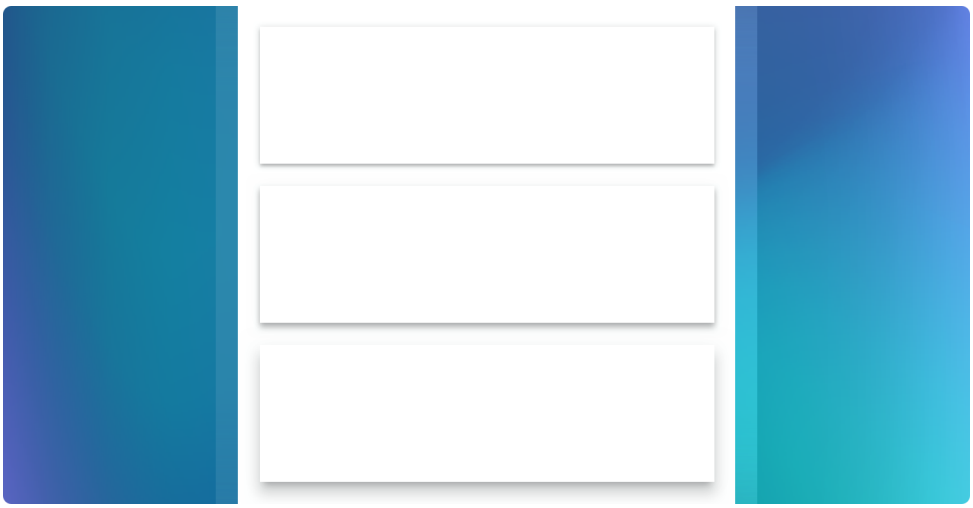
*Need help? Self-Service || Yammer || **No new spaces from Nov-23rd***

Surface

Created by Davyd Geyl, last modified by Genevieve Wang on 18 Feb 2021



Version

2

Availability

Web

Android

iOS

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Measured from the front of one material to the front of another, an element's surface indicates the distance between materials and the depth of its shadow. In a light environment, as an object becomes closer to the user in z-space the shadow becomes more diffused. In a dark environment, this is expressed through light i.e. the lightness of the surface area depicts closeness in z-space. Changes in shadow and colour depict depth.

Useful Docs

- Interaction States

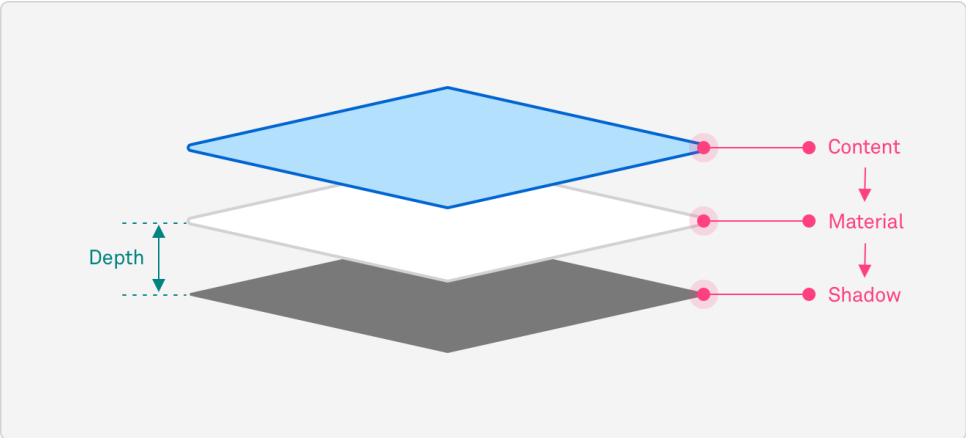
Specs

- Container & Interactive (Web)
- Container (Android & iOS)
- Interactive (Android)
- Interactive (iOS)

UI Playground

- Web Playground
- React Playground
- Android Catalogue app
- iOS Catalogue app

Terminology



Term	Definition
Content	Content appears on top of Materials and includes both Background and Foreground attributes.
Depth	Depth is the relative distance between two surfaces along the z-axis.
Material	The base colour that changes in a dark and light environment dependent on the depth.
Shadow	Shadow indicates distance dependent on the depth.
Basic Surface	The simple pattern that is a combination of material and shadow. Basic surfaces have no purpose attached. Basic surfaces are used to represent different states of Container and Interactive surfaces.
Container Surface	Container Surfaces are typically used as the background non-interactive UI elements that hold other UI elements.
Interactive Surface	Interactive Surface is when the entire surface is clickable or tappable as one UI element.

Usage (For Designers)

Shadows, Materials and Surfaces can be used to:

- Mimic real life objects and demonstrate visual hierarchy in digital space.
- Provide important visual cues about directional movement and provide a sense of depth.
- Indicate interactivity and provide affordance .

In short, Surfaces exist to break up content and provide section breaks to support user cognition.
To successfully depict a surface you must show:

1. Surface edges - this is shown by contrasting the surface from its surroundings.
2. Distance from other surfaces - this is depicted in partnership with light (shown with colour) and/or shadows - this relationship style is what makes up 'elevation'.
3. Overlap with other surfaces (either at rest or in motion) - shown with edges or distance.

For more details, see Google Material on Surfaces.

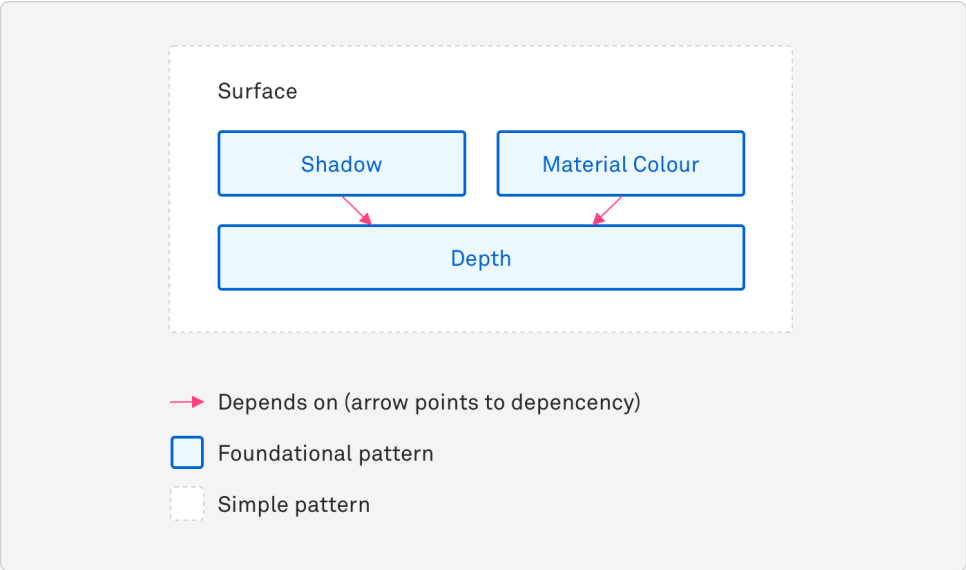
Usage (For Developers)

For web developers, see Usage for Surfaces in Able's web documentation.

For native developers, see iOS Catalogue and Android Catalogue for more examples.

Anatomy

The following is a dependency diagram which demonstrates how both *Shadow* and *Material colour* are dependent on *Depth*.
Attributes will change to indicate level in z-space.



Depth (Z-axis)

The range of imagined planes in the UI are on a scale from 0 to 24 with zero being the base and 24 the highest level of depth. The higher the level, the larger the shadow or lighter the surface will be.

- 0 - Base level
- 1
- 2
- 3
- 4
- 6
- 8
- 12
- 16
- 24 - Highest level

Shadow

Shadow indicates distance dependent on the depth.

For details on shadows used to represent each depth level, see spec pages below:

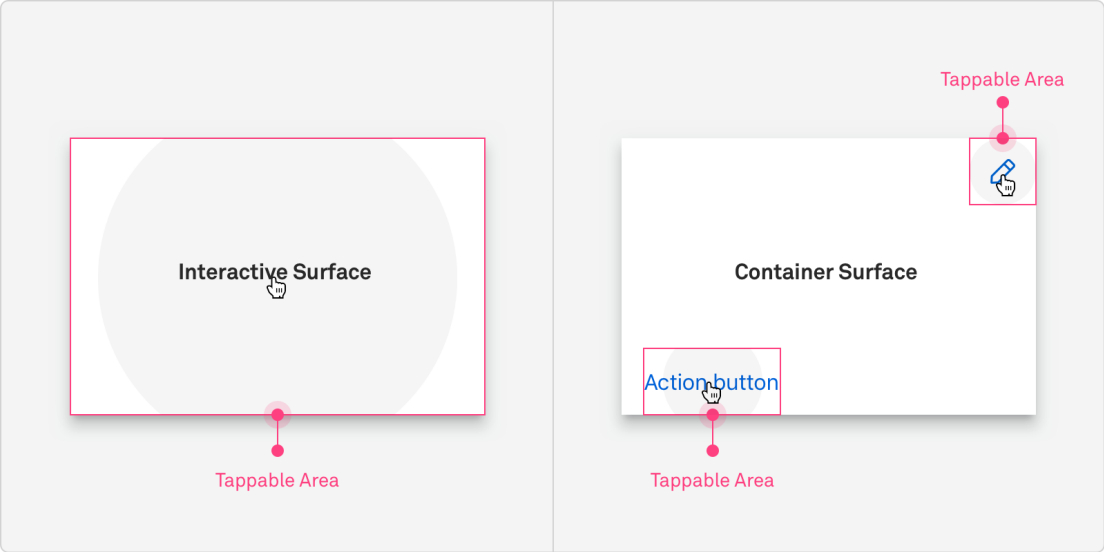
- Shadows Specs on Android
- Shadows Specs on iOS
- Shadows Specs on Web

Material

The full list of materials to represent depth is defined on the Material pages:

- Android & iOS Specs

Understanding Your Surface Interactivity



Raised content provides appropriate visual affordance to indicate they are interactive intuitively.

To ensure the user interaction is reflected appropriately, we defined 2 different surfaces based on their interaction design, they are:

1. Interactive Surface (Left)
2. Container Surface (Right)

Interactive Surface

An *Interactive Surface* is when the **entire surface is clickable as one** UI element. The surface will 'move down' on z-axis on press down interaction, or 'move up' on drag. Any movement along the z-axis is a result of a user interaction.

- An Interactive Surface must only have a single interaction (or click through) within it.
- No additional interactive UI element (e.g. List, Action Buttons, etc) should be added the surface. **for web, see note below.*
- Don't apply additional animation to the surface, such as rotation or fade.

> **Web specific instructions**
> If the content within the Surface is complex and has multiple text-nodes, we must introduce an Inline Link for accessibility. The link should be in context enough to pass 2.4.4 Link Purpose. The result must be reviewed with your team's Accessibility Manager.
>
> Please refer to Able web playground for code examples.

Container Surface

A *Container Surface* is when the **surface itself has no interactions**. It's designed to behave like a static element, hence, it does not move along the z-axis. An example of this would be a Card with an Action Button within it.
Container Surfaces are typically used as the background of a Card UI or Container UI, they hold all its child UI elements, and their size is determined by the space those elements occupy.

Keep in mind:

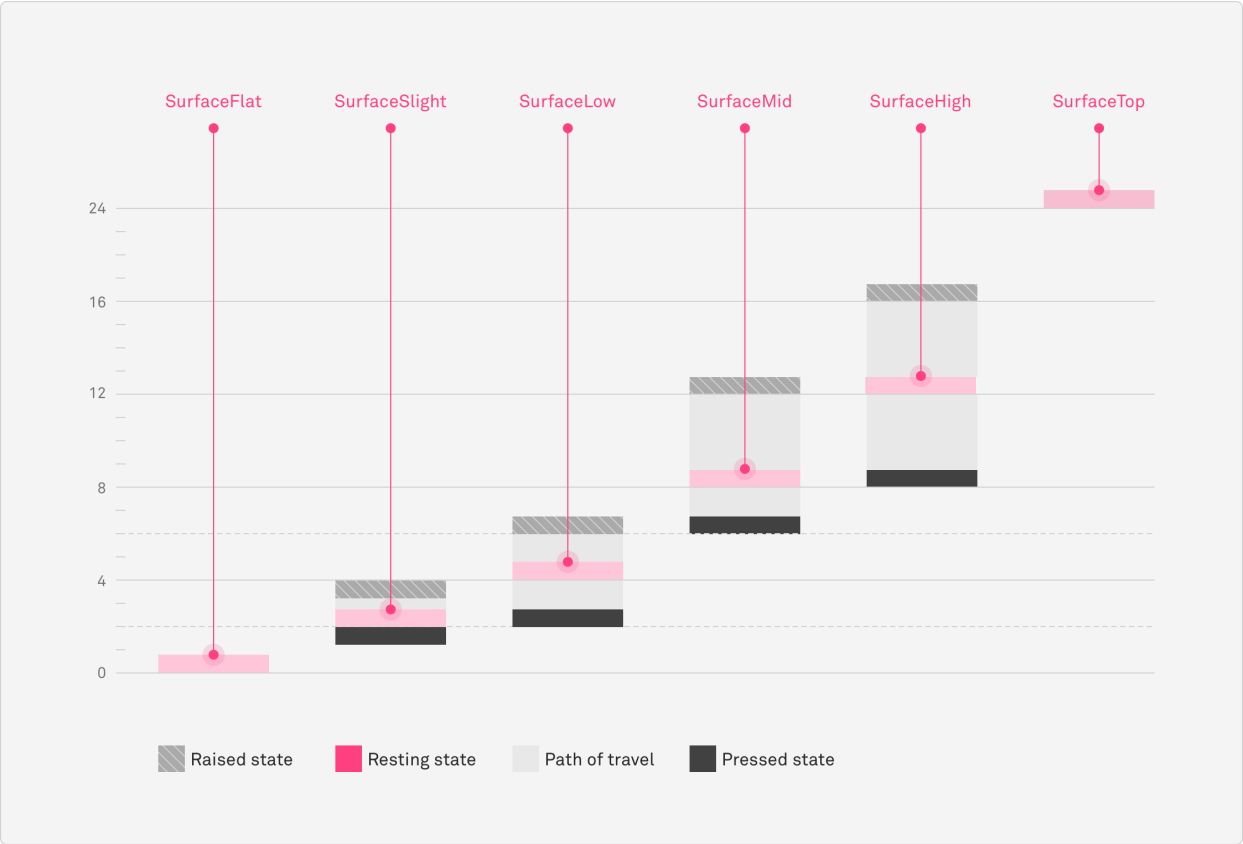
- A Container Surface must have at least 1 interactive control embedded within, as all raised content should be clickable in some way.
- It may have more than one interactive control in it, but be mindful of embedding too many within 1 surface.

It is recommended to mostly use Interactive Surfaces - i.e. the whole card is clickable driving a focused user interaction.
Container Surface may be used in patterns such as lists or cards where multiple interactions are required.

Surface Usage Recommendations

The following diagram shows the path of travel of each Surface. SurfaceFlat and SurfaceTop are only available at their Resting Elevation.
Resting Elevation refers to the default shadow and material for each pattern represented by one of the Basic Surfaces.

SurfaceSlight, SurfaceLow, SurfaceMid and SurfaceHigh are available as dynamic interactive surfaces with predefined interactive properties.
These levels can also be used as a Container Surface - in this case only the Resting Surface will be shown.



The following table lists Surface usage recommendations to ensure consistent application across digital touchpoints.
Certain components are positioned at higher levels than others to communicate hierarchy.

Interactive Surfaces

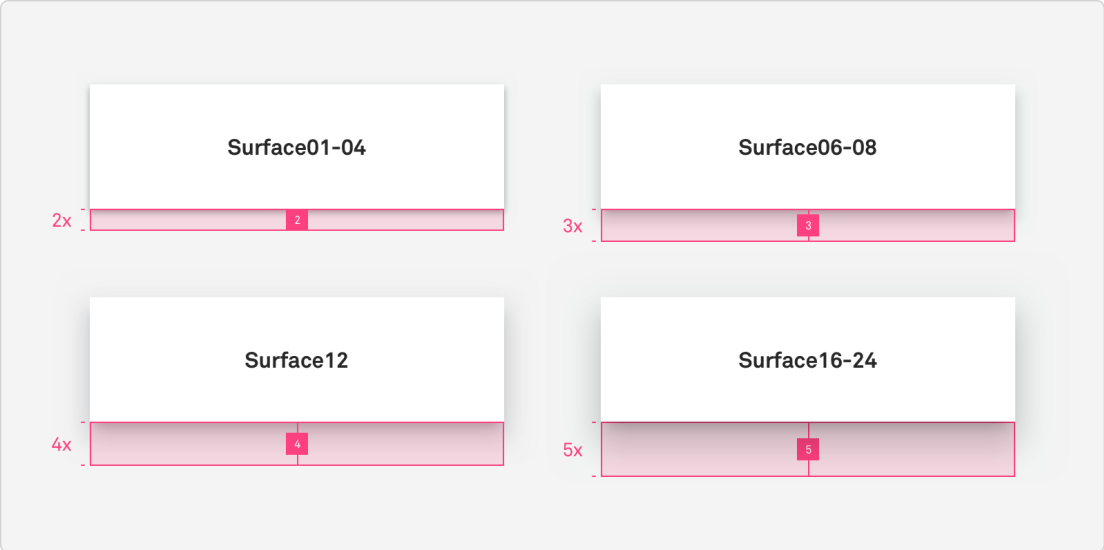
Pattern Name	Resting Surface		Interactive	Usage recommendations
	iOS, Android & Web XS+SM	Web MD+LG		
InteractiveSurfaceFlat	SurfaceBase		✅	Rows in flat Lists, such as in Settings
InteractiveSurfaceSlight	Surface02	Surface04	✅	Switch
InteractiveSurfaceLow	Surface04	Surface08	✅	Raised interactive Cards
InteractiveSurfaceMid	Surface08	Surface12	✅	Alerts, Toasts, Notifications, Tooltip, Promo Tiles
InteractiveSurfaceHigh	Surface12	Surface16	✅	Floating Action Button

Container Surfaces

Pattern Name	Surface		Interactive	Usage recommendations
	iOS, Android & Web XS+SM	Web MD+LG		
ContainerSurfaceFlat	SurfaceBase		❌	Screen Background, Page Content Background (e.g. for Text, Button etc), Lists, Bordered non-interactive Cards
ContainerSurfaceFlatGrey	SurfaceBaseGrey		❌	Secondary content background.
ContainerSurfaceFlatBlue	SurfaceBaseBlue		❌	Tertiary content background.
ContainerSurfaceSlight	Surface02	Surface04	❌	*TBD*
ContainerSurfaceLow	Surface04	Surface08	❌	Sticky Navigation, Bottom Nav
ContainerSurfaceMid	Surface08	Surface12	❌	Promo Tiles, Overflow Menu
ContainerSurfaceHigh	Surface12	Surface16	❌	Featured Product Tile
ContainerSurfaceTop	Surface24	Surface24	❌	Reserved for Modals / Dialogs *Requires 80% black background overlay behind Modal *No interactions beyond Resting Surface

Recommended Surface Margins

Below is a breakdown of the recommended minimum margin to allow for required shadow diffusion. Refer to Spacing for specs.
Please note: shadow24 is recommended to appear with 80% black overlay.



Surfaces	Recommended Margin
Surface01	spacing2x
Surface02	
Surface03	
Surface04	
Surface06	spacing3x
Surface08	
Surface12	spacing4x
Surface16	spacing5x
Surface24	

Best Practices (For Designers)

- **Consider a Surface a solid material**
  That means content and / or user input cannot pass through a surface.
  To successfully depict a surface you must show:
  - Surface edges - this is shown by contrasting the surface from its surroundings
  - Distance from other surfaces - this is depicted in partnership with light (shown with colour) and/or shadows - this relationship style what makes up ‘elevation’
  - Overlap with other surfaces (either at rest or in motion) - shown with edges or distance

- **Surfaces should reflect the physical world**
  This means a Surface can be stacked or attached to one another, but cannot pass through each other.
  Surfaces **should never** occupy the same z-depth or imagined plane in the same space.

- **Surfaces should contain Content**
  Content refers to Text, Colour, Images, Buttons etc - Content does not add thickness to a Surface.
  If a Surface is Pressed or Dragged Content must also also depress/lift accordingly.

- **A Surface that can be reordered must follow predefined 'UI physics'**
  If a user wishes to reorder content - for example a series of Cards - a Surface can only move along the X or Y axis i.e. up/down or left/right
  A Surface can not rotate or move along any plane.
  The pick up and move gesture allows a user to reorder cards within a collection.
  On selection a cards shadow increases to signify 'pick up'
  On move cards reshuffle to allow for reordering content into new space.

- **A Surface can be used progressively reveal content**
  A Surface may be stretchable meaning a Surface can grow or shrink along one or more edges up to a size limit to expand/collapse or show/hide content.

- **A Surface can not change opacity**
  The opacity of a Surface has been pre-defined and must not be altered.

- **Surfaces can be represented by colour, raised elements or a combination of both**
  As long as a Surface has a clearly defined edge, a Surface can occupy a combination of planes along the z-axis.

Accessibility Considerations

Surfaces are designed to meet the WCAG 2.1 Success Criteria outlined below. All requirements should be considered and understood when using the pattern. Requirements marked with ⚠️ indicate partial compliance or compliance must be met in build.

All Surfaces

- 1.4.4 Resize Text (AA) - When resized to 200%, text is not truncated
- 1.4.10 Reflow (AA) - When scaled to 400%, there is no loss of content or functionality

Interactive Surface Only

- 2.1.1 Keyboard (A) - Interactive surfaces are accessible by keyboard
- 2.4.3 Focus Order (A) - Interactive surfaces receive focus and should be present in a meaningful focus order sequence.
  ⚠️ Partial requirement met - ensure logical order when in use
- 2.4.7 Focus Visible (AA) - Interactive surface have a visible focus state.
- 4.1.2 Name, Role, Value (A) - Interactive surfaces have a name and role that can be programmatically determined

Keyboard

Users must be able to navigate to and select an interactive surface using the keyboard alone. For web, depending on whether a surface is implemented as a link or button, interactions will be slightly different.

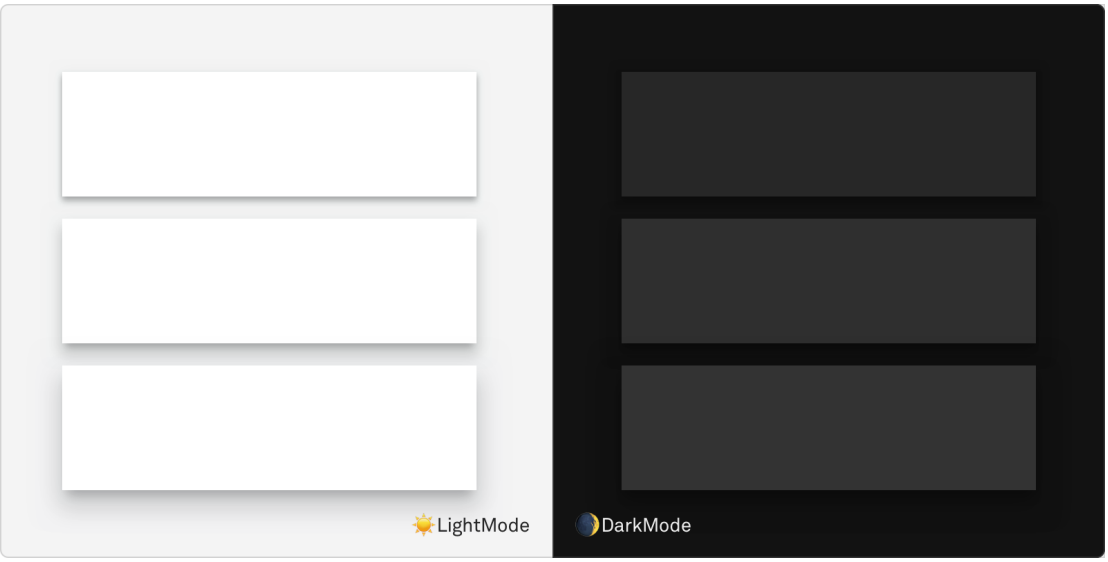
Screen Reader

Contents within surfaces form a natural group, so consideration should be given to present this information to screen reader users. Grouping these elements in a container that receives screen reader focus will allow accessiblity services to present inner elements descriptions in a single announcement one after the other. This consolidation of related elements helps user of assistive technology discover the information on the screen more efficiently.

For native app, interactive surfaces must have a button role applied to the contents within. Screen readers will announce the grouped contents of the interactive surface along with "button".

On web, interactive surfaces may be implemented as a link <a> or button <button>.

Capabilities



Surfaces will automatically update its appearance based on user's light or dark appearance settings.

No labels